

Judge Q&A Preparation — PS 26083

Every answer below traces back to a Decision Log entry. If a judge asks something not covered here, the honest fallback is: "Good question — that's not something we've locked yet; here's our current thinking and what would resolve it" — never invent an answer on the spot.

Q: Why UTCI, and not just Heat Index or WBGT?

A: UTCI integrates air temperature, humidity, wind speed, AND mean radiant temperature (solar/thermal radiation) — the PS itself calls out that dry-bulb-only warnings miss humidity/wind/radiation compounding effects, and UTCI is the most complete standard index that captures all of them together. WBGT needs a separate natural-wet-bulb/black-globe derivation path that would have consumed real build time without adding much beyond what UTCI already gives us, so we kept it as a secondary comparison metric only. *(Decision Log D3)*

Q: What is MRT and why does it matter?

A: Mean Radiant Temperature — the temperature the human body "feels" from radiant heat sources (sun, hot surfaces) around it, separate from air temperature. We compute it from ERA5's radiation variables (direct/diffuse solar, thermal radiation) using the established ERA5-HEAT/thermofeel method — not an invented shortcut. It's the input that lets UTCI capture "standing in the sun" vs "standing in shade" even at the same air temperature. *(Decision Log D4)*

Q: ERA5 is a ~25km grid. How do you get ward-level results from that?

A: We're honest that ERA5 itself is NOT ward-level — we never claim that. We spatially join each of the 9 ERA5 grid points covering Ahmedabad to the nearest ward centroid (documented nearest-grid-point method, not exact interpolation). This gives every ward a

real, physically-grounded thermal value, but we're explicit that it's a spatial approximation, not native ward-resolution weather data. (*Decision Log, "what we cannot claim": "ERA5 gives ward-level weather" — explicitly listed as something we do NOT say*)

Q: What does NCTL actually prove, scientifically?

A: NCTL (Nighttime Cumulative Thermal Load) quantifies something conventional heat alerts ignore: how much a location's nighttime temperature stays above its OWN historical baseline (90th percentile of local nighttime UTCI), summed in degree-hours. It doesn't "prove" health outcomes on its own — we're explicit it's not a clinically validated recovery index — but it's a real, reproducible, physically-grounded signal that heat-health literature links to reduced overnight recovery during heatwaves, which conventional single-day alerts completely miss. (*Decision Log D5*)

Q: Why $\alpha = 0.7$ for the CTBI recurrence? Is that scientifically correct?

A: We do NOT claim 0.7 is "the" scientifically correct decay factor — that's explicitly on our "cannot claim" list. We tested $\alpha \in \{0.6, 0.7, 0.8\}$ and the qualitative pattern (CTBI rises to a seasonal peak, then declines) is stable across all three. We locked 0.7 as the default because it's the only value where $\alpha + \beta + \gamma$ sums to exactly 1.0 ($\beta = \gamma = 0.15$), giving a mathematically clean weighted average — a practical, defensible choice, not a claim of scientific optimality. Once real health-outcome data exists, these weights could be fit by regression instead. (*Decision Log D6, D6a*)

Q: Where's your mortality/health-impact model?

A: Deliberately not built, and we say so directly: reliable ward-level hospitalization/mortality labels aren't publicly available at the granularity we'd need. We chose not to fabricate training labels just to have an ML model to show. Our current output is an interpretable, rule-based priority proxy (CTBI + vulnerability + response capacity → Low/Moderate/High/Critical), not a trained predictive model. The moment real health-outcome data becomes available, this exact feature set (CTBI, NCTL, vulnerability,

response capacity) becomes direct input to a supervised model — architecture is ready for it, data isn't. (*Decision Log, Phase 2 / Future roadmap*)

Q: Where's your SHAP / explainability analysis?

A: None run — and we don't fake one. SHAP explains a trained model's predictions; we don't have a trained health-outcome model yet (see above), so there's nothing genuine for SHAP to explain. Any SHAP-style graphic not from a real trained model is explicitly labeled "Illustrative" in our maturity framework — we chose not to include even that, to avoid any ambiguity about what's real. Our "Explain" step today (see the Predict→Explain→Prioritize→Act slide) is transparent rule-based reasoning — which ward, which drivers — not model-explainability in the SHAP sense.

Q: How is this ward-level if your vulnerability data mismatch happened?

A: We initially tried the 2011 Census (real, publicly available, ward-level data) but discovered it uses AMC's OLD ward delimitation — 57 core wards + 6 outgrowth areas, vs the CURRENT 48-ward structure (redrawn ~2021-2022 for women's-reservation compliance). No official crosswalk between the two exists, so a direct join would have silently attached the wrong ward's population/literacy data to the wrong current ward — a serious, hard-to-detect integrity risk. We rejected that source entirely rather than force-fit it. Instead we used WorldPop (population density) and Sentinel-2 NDVI (green cover) — both RASTER datasets that aggregate directly onto our CURRENT 48-ward polygons with zero mismatch risk. Sanity-checked against known city totals (population, area) and known geography (dense old-core wards vs leafier outer wards) — both checks passed. (*Decision Log D7, D8*)

Q: Why only 2 vulnerability features (population density, green cover)? What about elderly%, outdoor workers, slums?

A: No reliable, ward-matched public source exists for those within our timeframe. Following our own rule — "3 real features beat 10 invented ones" — we chose to run on fewer, verifiable, satellite/census-derived features rather than fabricate demographic

percentages we couldn't source. This is documented as an open item in the Decision Log, not hidden.

Q: Did you actually send a WhatsApp/SMS alert?

A: No — and we don't claim we did. The action engine produces a real, structured recommendation payload per ward (grounded in Ahmedabad's own real Heat Action Plan escalation actions — cooling centers, outdoor-work-hour shifts, hospital alerts). Live SMS/WhatsApp delivery is an API stub, explicitly labeled as such, not a working integration — per our own priority ladder, this was always meant to be the last thing built, kept only if time allowed.

Q: Is your CTBI map showing real spatial variation, or is it the same everywhere?

A: Two honest layers here. Early in the build, CTBI itself was a single CITY-WIDE time series (not computed per grid-point), so a snapshot on any one date showed all 48 wards in the same thermal tier — real time variation (April→May→June), but no ward-to-ward variation yet. That was documented as a known limitation, not hidden as a "flat" bug. Once real ward-level vulnerability was integrated (population density + NDVI, D8/D10), the PRIORITY layer gained genuine ward-to-ward variation — 23 wards High, 25 Moderate, matching real Ahmedabad geography (dense old-core wards score higher than leafier outer wards). We can show both stages honestly if asked.

Q: What's your validation plan, if you don't have real accuracy numbers yet?

A: AUROC, F1, Recall, and calibration curves against real hospitalization/mortality records — once real institutional health data is accessible (Phase 2). We don't report fabricated accuracy numbers today because we have no ground truth to validate against yet; reporting a number without ground truth would be worse than reporting none.

Q: What's actually novel here vs. existing heat-warning systems?

A: Operationalizing persistent NIGHTTIME thermal load (NCTL) and multi-day cumulative thermal burden (CTBI) within a ward-level heat-health decision pipeline, then combining that with real vulnerability and response-capacity signals to move from "prediction" to "intervention priority" — not inventing UTCI, WBGT, or the concept of heatwave alerts themselves (we don't claim that).

Quick-reference: what's / / today

 Demonstrated	 Prototype	 Future
ERA5→UTCI→NCTL→CTBI pipeline, real 6-season data	Priority rubric (rule-based, capped below CRITICAL)	Health-outcome-trained model
48 real ward boundaries + spatial join	Action Engine (HAP-grounded, but not field-validated)	Real SHAP explainability
Vulnerability (WorldPop + NDVI), sanity-checked	Dashboard (real data, basic UI)	Live SMS/WhatsApp delivery
		Calibrated 3-5 day forecast (currently historical replay)